Cover Letter

November 13, 2026

Faculty Search Committee
David Crawford School of Engineering
Norwich University
Northfield, Vermont

Re: Assistant Professor of Electrical and Computer Engineering — F00092

Dear Members of the Search Committee:

I am applying for the tenure-track Assistant Professor position in Electrical and Computer Engineering at Norwich University, beginning July 2027. I earned my PhD in Electrical and Computer Engineering from the University of Oklahoma in 2025, specializing in AI and machine learning for reliable and resilient wireless networks. My background combines undergraduate university teaching, laboratory support, student research mentoring, peer-reviewed scholarship, and applied engineering experience at AT&T, AT&T Labs, Futurewei, and Impac Exploration Services. Norwich's emphasis on experiential education, leadership, and close faculty-student engagement is particularly appealing to me.

As a Lecturer at the University of Lahore, I taught Computer Programming, Communication Systems, and Wireless Communication and developed instructional materials, assessments, and student projects. I supported communications laboratory instruction at UET Taxila and mentored student researchers at COMSATS University Islamabad and the University of Oklahoma. I would contribute most immediately to programming, communications, signals and systems, digital signal processing, computer networking, and applied AI/ML. I would also prepare adjacent foundational courses, including introductory digital systems and control systems, in consultation with faculty and the available laboratory resources. My approach emphasizes connecting analytical principles with implementation, testing, and clear technical communication.

My research investigates reliable learning when data are sparse, noisy, or drawn from changing operating conditions. My doctoral work on wireless propagation modeling, generative and transfer learning, network outage detection, and resilience has led to peer-reviewed publications. Related research in wearable inertial-sensor activity recognition provides a foundation for intelligent sensing and trustworthy time-series analysis. In my current ML engineering assignment supporting AT&T's E911 service assurance, a root-cause classification workflow improved automated classification performance from approximately 20% to 80%. These experiences motivate a focused undergraduate research program on data-efficient learning, intelligent sensing, and resilient networked systems.

At Norwich, I would involve undergraduate researchers and senior design teams in achievable projects such as detecting anomalies in simulated network measurements, comparing sensor classifiers under changing conditions, and evaluating reliable responses to communication or sensing failures. I would welcome interdisciplinary collaboration with cybersecurity and embedded-systems colleagues to introduce security-aware design and failure analysis into appropriate ECE projects, without assuming specialized hardware-security expertise. Projects would use public or synthetic data and available laboratory equipment, with clear milestones, testing criteria, and opportunities for student presentations and publications.

I am also interested in student advising, career preparation, curriculum development, and ABET-oriented continuous improvement through documented learning outcomes, assessment rubrics, and evidence-based course revisions. I would bring industry-informed examples to the classroom and support students in developing technical judgment, teamwork, and professional responsibility. I would welcome the opportunity to contribute to Norwich's distinctive undergraduate engineering community.

Thank you for considering my application.

Sincerely,

Waseem Raza, PhD